**ALSTOM • IXL ENGINEERING SOLUTIONS**

# IXL Team Hours Tracker & Allocation System

## Comprehensive Technical Architecture, System Internals, & Developer Manual

**Version:** 2.0.0 (Pure Technical) **Target:** Software Engineers & AI Agents **Production:** ixl-hours-tracker.web.app **Git:** main branch

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> **Developer Maintenance Contract:** This document represents the authoritative single-source-of-truth technical blueprint for the entire application. Whenever any functionality, data structure, or endpoint is added, modified, or removed, this document MUST be updated accordingly.

## 1. Executive System Overview

The **Alstom IXL Team Hours Tracker** is a specialized high-performance Single Page Application (SPA) designed to track, schedule, and reconcile daily engineering hours for multi-regional railway signaling teams in the **Cairo** and **Istanbul** engineering offices.

Key architectural goals:

- **Double-Entry Accounting Reconciliation:** Project hours aggregated across the system strictly match individual team member hours at all times (Project Sum = Employee Sum).
- **Dual Office Calendars:** Cairo (Friday/Saturday weekend) vs. Istanbul (Saturday/Sunday weekend) automated compliance.
- **Deterministic Zero-Framework Architecture:** Vanilla ES6+, Semantic HTML5, and CSS3 with hardware-accelerated touch physics. Zero heavy build tools or framework overhead.
- **Zero-Emoji Standard:** Strict corporate constraint prohibiting Unicode emojis across UI text, code symbols, test suites, and commit logs.

```
[Client Web Browser]
   ├── Desktop (PC)    --> Excel-Style Grid, 2D Multi-Select, Keyboard Navigation
   └── Mobile Touch    --> Mobile Editor (Form/Cards), Agenda List, 1-Tap Starred Chips
        |
        ├── In-Memory State Machine: window.state
        ├── Local Storage & Cookie Fallback Cache (Synchronous optimistic updates)
        └── Network REST Sync: POST /api/sync (Atomic payload with SHA-256 validation)
             |
             ├── Node.js Server (server.js) OR PowerShell Server (server.ps1)
```

```
                └── Google Firebase Hosting Edge CDN (Production)
```

## 2. Complete File Inventory & Code Map

| File Path | Size | Architectural Role & Description |
|---|---|---|
| `index.html` | 109 KB | Complete SPA markup shell: schedule table, mobile cards, popovers, modals, responsive pictures. |
| `app.js` | 368 KB | Master client state engine (~8,800 lines): DOM event delegation, math reconciliation, state sync, identity persistence. |
| `styles.css` | 141 KB | Design system (~6,100 lines): CSS variables, light/dark themes, responsive layout breakpoints, dual-axis touch physics. |
| `data.json` | 20 KB | Master single-source-of-truth database: sections, employees, projects, allocations, admin PIN hash, audit log. |
| `data.backup.json` | 14 KB | Cold-start fallback image used for automatic recovery by local servers. |
| `server.js` | 3.8 KB | Node.js Express REST API server providing `GET /api/data` and `POST /api/sync`. |
| `server.ps1` | 8.5 KB | Native Windows PowerShell HTTP listener for restricted enterprise systems lacking Node.js runtime. |
| `firebase.json` | 675 B | Firebase Hosting configuration: route rewrites, CDN cache-control headers, deploy ignore rules. |
| `cloudflared.exe` | 55.3 MB | Cloudflare Tunnel binary for instant public link generation via `Share-Public-Link.bat`. |
| `test_*.js & *.ps1` | ~200 KB | 12 automated test suites verifying 2,150+ unit, integration, and security assertions. |

# 3. Master Data Models & State Architecture

## 3.1 The In-Memory State Object ( `window.state` )

```
window.state = {
  sections: [],         // Office sections: Cairo, Istanbul
  employees: [],        // 26 Team members (active & deactivated)
  projects: [],         // 55 Projects, tenders, training & support
  allocations: {},      // Hash map: "empId_YYYY-MM-DD" -> { project, hours }
  auditHistory: [],     // Chronological security and edit events
  adminPasswordHash: "",// SHA-256 hash for Peter MOUNIR admin mode
  currentUserId: null,  // Active user profile ID ("emp_1", ...)
  isAdmin: false,       // Master admin privilege flag
  isViewOnly: false,    // Read-only URL parameter flag (?mode=view)
  currentYear: 2026,    // Active calendar year (starts Nov 2026)
  currentMonth: 11,     // Active calendar month (1-based: 11 = November)
  deviceMode: "pc",     // "pc" or "mobile"
  mobileViewMode: "form", // "form" (Editor) or "grid" (Viewer)
  selectedCells: new Set(), // Set of active composite keys in 2D multi-select
  activeKeyboardCell: null  // Composite key holding active keyboard focus
};
```

## 3.2 Composite Key Architecture

Allocations are keyed by composite strings formatted as `${employeeId}_${YYYY-MM-DD}` (e.g. `emp_22_2026-11-04` ).

To prevent string splitting bugs when employee IDs contain underscores (such as `emp_22` ), keys are parsed using `lastIndexOf('_')` :

```
const lastUnderscore = key.lastIndexOf('_');
const empId = key.substring(0, lastUnderscore);
const dateStr = key.substring(lastUnderscore + 1);
```

# 4. Dual Office Calendar & Weekend Rules

| Office Section | Location | Official Weekend (Holidays) | Official Workdays |
|---|---|---|---|
| **Applications - Cairo** | Cairo | Friday (5), Saturday (6) | Sunday through Thursday |
| **ILS - Istanbul** | Istanbul | Saturday (6), Sunday (0) | Monday through Friday |

### Multi-Day Range vs. Single Weekend Overtime Rule

- **Multi-Day Range:** When filling a date range that spans a weekend, the chosen holiday option is applied:
  - `dayoff`: Weekends are set to `DAY OFF` with 0h. Workdays get the project.
  - `include`: Weekends are worked and filled with project and hours.
  - `skip`: Weekends are untouched.
- **Explicit Single Weekend Day:** If an engineer clicks a single weekend day (e.g. Friday in Cairo), the system permits direct overtime logging without forcing `DAY OFF`.

## 5. Desktop (PC) Interface Architecture

- **Excel-Style Grid Navigation:** Arrow keys move the focused cell across days and employees. `Enter` opens the popover. `Delete` clears allocations.
- **Pointer State Machine:**
  - Single Click: Stands on cell without opening popovers (cleans keyboard focus).
  - Double Click: Opens the Project Selection Popover.
  - 380ms Long Press: Enters 2D rectangular multi-cell selection.
  - 8px Jitter Deadband: Prevents micro-mouse tremors from accidentally entering drag mode.
- **Admin 2D Multi-Selection:** In Peter MOUNIR's Admin Mode, multi-cell drag expands across multiple employee rows and dates simultaneously.

## 6. Mobile Interface Architecture & Touch Physics

- **Dual Modes:** Mobile Editor mode (cards, agenda, 1-tap starred chips) vs. Read-Only Grid mode (horizontal table view).
- **Dual-Axis Touch Physics:** All table wrappers enforce `touch-action: pan-x pan-y !important` and `overflow-x: auto !important`, ensuring effortless left/right swipe navigation on phone displays.
- **Add Member Form Baseline Alignment:** `#addMemberBox .form-grid-3` enforces `align-items: flex-end` with `min-height: 28px` label containers so Full Name, Section, and Default Daily Hours inputs share an exact horizontal baseline.
- **Profile-Isolated Starred Projects:** Stored per-user in `localStorage` under `ixl_starred_{userId}`. 1-tap logging chips appear at the top of the mobile logger.

## 7. Role-Based Access Control & View-Only Defense

| Role | Identification | Capabilities & Boundaries |
|---|---|---|
| **View-Only** | `?mode=view` URL query | Read-only inspection. All buttons, popovers, drawers, and sync requests are hard-blocked via CSS and JS guards. |
| **Team Member** | Chosen identity (e.g. Sarah KHALED) | Can edit only their own row. All other team member rows are read-only. |
| **Master Admin** | Peter MOUNIR (PIN-authenticated) | Full global rights: edit any row, 2D drag across team, add/deactivate members, move sections, adjust project baselines. |

## 8. Automated Test Harness (12 Suites / 2,150+ Assertions)

| Test Suite | Tests | Core Verifications |
|---|---|---|
| `test_all_users_integrity.js` | 1,822 | Row edit permission isolation across all 26 users; popovers, keyboard delete, audit events. |
| `test_suite.ps1` | 92 | Database schema, Cairo/Istanbul rules, timezone rollover, DOM elements, PIN hashing. |
| `test_starred_projects_integrity.js` | 52 | Profile isolation of starred favorites, 1-tap chips, star protection for special items. |
| `test_cell_click_hold_sequence.js` | 47 | 380ms hold timer, 8px micro-jitter deadband, standing focus, arrow navigation. |
| `test_calculation_integrity.ps1` | 39 | Double-entry accounting balance, baseline hours arithmetic, 1h-12h boundary clamps. |
| `test_weekend_hours_assignment.js` | 28 | Cairo vs. Istanbul weekend range logic vs. single-day overtime assignments. |
| `test_pc_view_only_integrity.js` | 27 | PC read-only URL defense: DOM button removal, popover blocking, key blocking. |
| `test_mobile_view_only_integrity.js` | 27 | Mobile read-only URL defense: grid lock, form hiding, apply mutation rejection. |
| `test_mobile_project_browser.js` | 25 | Mobile logger agenda selection, integer hour stepping (1-12h), special project naming. |
| `test_mobile_list_scroll_alignment.js` | 23 | Mobile table minimum widths, 2-axis touch physics, Add Member form baseline alignment. |
| `test_pc_bulk_clear_integrity.js` | 15 | Bulk fill clear action normalization, star immunity, Projects List chooser. |
| `test_pc_welcome_screen_integrity.js` | 11 | PC and mobile welcome splash full-bleed screen fit, spinner placement, asset parity. |

## 9. Developer Guidelines & Ongoing Maintenance Contract

1. **Mandatory Document Update:** Whenever you add, modify, or delete any feature, data model, or style rule, you MUST immediately update both `TECHNICAL_ARCHITECTURE.md` and `TECHNICAL_ARCHITECTURE.html`.
2. **Strict Zero-Emoji Policy:** Never use Unicode emojis in code, commits, or documentation.
3. **Preserve Double-Entry Math:** Always run `test_calculation_integrity.ps1` after altering hours calculation logic.
4. **CRLF Line Endings:** Preserve Windows CRLF ( `\r\n` ) line endings across repository files.
5. **Timezone Safety:** Always construct dates via `parseLocalDate(dateStr)`, never using raw UTC midnight strings.
6. **Cache Busting:** When updating `styles.css` or `app.js`, bump the cache query parameter in `index.html` (e.g. `?v=10.0`).